Apex Planet Internship – Task 3

Deep-Dive Analysis & Interactive Dashboarding

1. Project Overview

This project analyzes sales transaction data to identify revenue trends, product performance, and customer-segment patterns. An interactive Power BI dashboard was developed to help users explore performance through KPI cards, charts, and slicers.

2. Business Objectives

- Monitor core sales performance KPIs.
- Identify high-revenue products and categories.
- Analyze monthly sales patterns.
- Compare sales across customer age groups.
- Enable interactive exploration of the sales data.

3. Dataset

The analysis uses the cleaned sales dataset prepared during Task 1. The dataset includes order dates, customer attributes, products, categories, quantities, unit prices, and total sales.

4. Core KPIs

- Total Sales = Sum of Total_Sales.
- Transaction Records = Number of rows in the dataset.
- Total Units Sold = Sum of Quantity.
- Average Sales per Transaction Record = Total Sales / Transaction Records.
- Average Selling Price = Total Sales / Total Units Sold.

5. Deep-Dive Methodology

The analysis examines monthly sales trends, product revenue, category revenue, and sales by customer age group. Power BI slicers allow users to filter the report by date, city, gender, product category, product, and age group.

6. Key Findings

Insert verified findings from the completed dashboard:

- Highest-revenue product: [Laptop].
- Highest-revenue category: [Electronics].

- Highest-revenue age group: [35-44].
- Best-performing month: [November].

7. Business Recommendations

Use the verified results to recommend areas for further investigation, such as reviewing high-revenue products, understanding differences between customer segments, and monitoring monthly performance.

8. Dashboard Features

The dashboard includes five KPI cards, four analytical charts, and interactive slicers. These allow users to explore sales performance without manually filtering the source data.

9. Limitations

The dataset contains sales transactions but does not include product costs, marketing spend, website visits, or customer churn status. Therefore, profit, marketing ROI, conversion rate, and churn cannot be calculated reliably from this dataset alone.

10. Conclusion

This project demonstrates a practical workflow combining KPI development, sales analysis, customer segmentation, and interactive business intelligence reporting using Power BI.